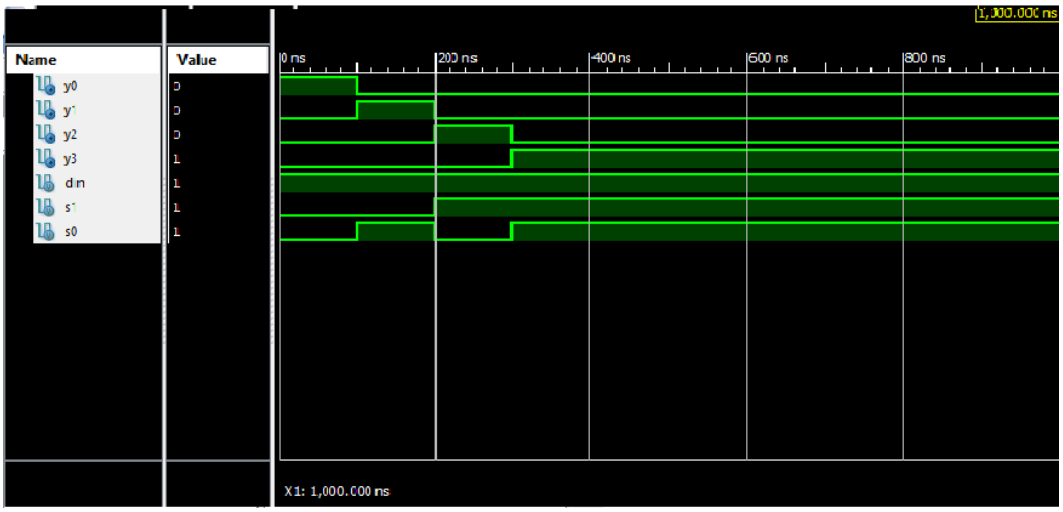
DEMUX 1:4

```
module jay(
   input din,
   input s1, s0,
   output reg y0, y1, y2, y3
);
always @(*) begin
   y0 = 0;
   y1 = 0;
   y2 = 0;
   y3 = 0;
   if (s1 == 0 && s0 == 0)
      y0 = din;
   else if (s1 == 0 && s0 == 1)
      y1 = din;
   else if (s1 == 1 && s0 == 0)
      y2 = din;
   else
      y3 = din;
end
endmodule
```

OUTPUT



VERILOG TEST FIXTURE

```
module HNN;
        reg din;
        reg s1;
        reg s0;
        wire y0;
        wire y1;
        wire y2;
        wire y3;
        jay uut (
                .din(din),
                .s1(s1),
                .s0(s0),
                .y0(y0),
                .y1(y1),
                .y2(y2),
                .y3(y3)
        );
```

```
        initial begin
                // Initialize Inputs
                din = 1;
                s1 = 0;
                s0 = 0;
                #100;
                din = 1;
                s1 = 0;
                s0 = 1;
                #100;
                din = 1;
                s1 = 1;
                s0 = 0;
                #100;
                din = 1;
                s1 = 1;
                s0 = 1;
                #100;
        end
endmodule
```